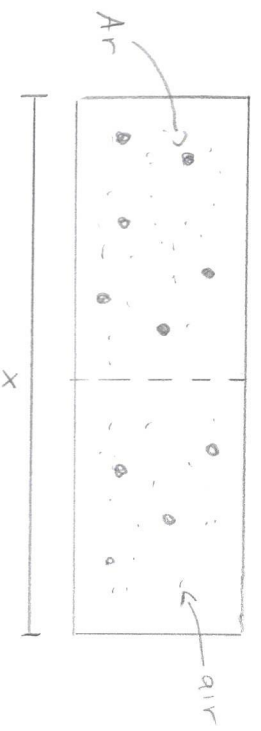


Transport

$\vec{J}_x$



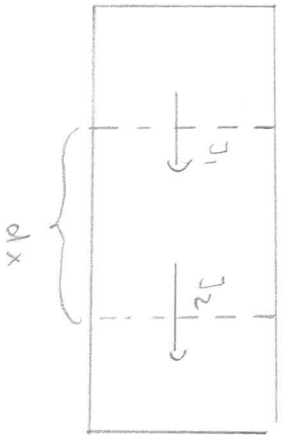
Diffusion of Argon in air

$$flux_x = \underbrace{-D}_{\text{diffusion coefficient}} \underbrace{\frac{\partial c}{\partial x}}_{\text{concentration gradient}}$$

Fick's first law

$$\left[\frac{\#}{m^2 s} \right] \left[\frac{m^2}{s} \right]$$

Concentration c is a function of time and location $c(x, t)$



$$\frac{\partial c(x, t)}{\partial t} = - \underbrace{\frac{(j_2 - j_1) A}{\underbrace{dx}_{\text{Volume}}}}_{\text{flux difference}}$$

$$\frac{\partial c(x, t)}{\partial t} = - \frac{\partial j}{\partial x} = \frac{\partial}{\partial x} \frac{\partial c}{\partial t}$$

3D: $c(x, y, z, t)$

$$\frac{\partial c}{\partial t} - \frac{\partial}{\partial x} \frac{\partial c}{\partial x} - \frac{\partial}{\partial y} \frac{\partial c}{\partial y} - \frac{\partial}{\partial z} \frac{\partial c}{\partial z} = 0$$

$$\frac{\partial c}{\partial t} - \nabla \cdot (D \nabla c) = 0$$

$$\frac{\partial c}{\partial t} - \nabla \cdot (D \nabla c) = S \quad \begin{matrix} \text{internal} \\ \text{sources or sinks} \end{matrix}$$

Advection: transport by flow

$$\uparrow$$

$$Dc(x, y, z, t) = \frac{\partial c}{\partial x} dx + \frac{\partial c}{\partial y} dy + \frac{\partial c}{\partial z} dz + \frac{\partial c}{\partial t} dt$$

total derivative

$$\frac{Dc}{dt} = \underbrace{\frac{\partial c}{\partial x} \frac{dx}{dt}}_{u_x} + \underbrace{\frac{\partial c}{\partial y} \frac{dy}{dt}}_{u_y} + \underbrace{\frac{\partial c}{\partial z} \frac{dz}{dt}}_{u_z} + \frac{\partial c}{\partial t}$$

$$\underbrace{(\vec{u} \cdot \nabla)}_{\text{advection operator}} = [u_x, u_y, u_z] \cdot \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right]$$

$$\vec{u} = [u_x, u_y, u_z]$$

flow vector

$$= \frac{\partial c}{\partial t} + (\vec{u} \cdot \nabla) c$$

more general

$$= \frac{\partial c}{\partial t} + \underbrace{\nabla \cdot (\vec{u} c)}_{\vec{u} \cdot (\nabla c) + (\nabla c) \cdot \vec{u}} + (\vec{u} \cdot \nabla) c + c (\nabla \cdot \vec{u})$$

"product rule"

$$\nabla \cdot \vec{u} = 0 \quad \text{for incompressible flow}$$

advection - diffusion equation

$$\underbrace{\frac{\partial c}{\partial t}}_{\text{local change}} - \underbrace{\nabla \cdot (D \nabla c)}_{\text{diffusion}} + \underbrace{\nabla \cdot (c \vec{u})}_{\text{advection}} = \underbrace{S}_{\text{sources - sinks}}$$

Heat equation

$$\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T) + \nabla \cdot (c \vec{u} T) = \underbrace{q}_{\text{heat added / subtracted to unit volume}}$$

$$\frac{\text{J}}{\text{s m}^3}$$

$$\frac{\text{J}}{\text{s m}^3}$$

Basic 1D heat equation

- no advection ($\nabla \cdot (\vec{u} T) = 0$)
- no sources / sinks ($q = 0$)
- 1 dimension ($\frac{\partial T}{\partial y} = 0$ $\frac{\partial T}{\partial z} = 0$)
- $k, \rho, c_p = \text{constant}$

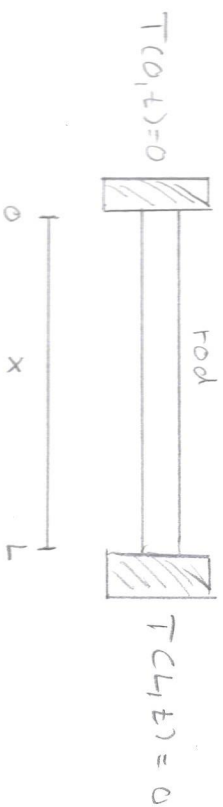
$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right)$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

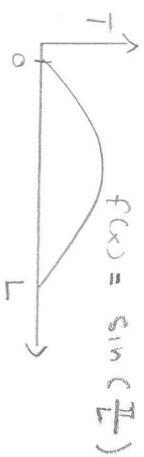
$$\alpha = \frac{k}{\rho c_p} \quad \left[\frac{\text{m}^2}{\text{s}} \right]$$

"thermal diffusivity"

Solving PDEs: Separation of variables



$$T(x, 0) = f(x)$$



$$\text{PDE: } \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Problem: Seek $T(x, t)$ that satisfies initial and boundary conditions

Approach: Separation of variables (functions)

- works for linear + homogeneous systems

- assume that solutions will be a product of two functions each function only with one variable

$$T(x, t) = A(x) B(t)$$

- apply proposed solution to PDE

$$\frac{\partial(A(x)B(t))}{\partial t} = \alpha \frac{\partial^2(A(x)B(t))}{\partial x^2}$$

$$A \frac{dB}{dt} = \alpha B \frac{d^2A}{dx^2}$$

- separate functions

$$\underbrace{\frac{1}{\alpha B} \frac{dB}{dt}}_{\text{function of } t \text{ only}} = \underbrace{\frac{1}{A} \frac{d^2A}{dx^2}}_{\text{function of } x \text{ only}} = -m$$

makes solution easier later

must be equal for all t and x values

$\Rightarrow$ must equal to a const.

time ODE

$$\frac{dB}{dt} = -\alpha m B$$

→ two ordinary diff Eq.

space ODE

$$\frac{d^2 A}{dx^2} = -m A$$

- solve space ODE

$$T(x, t) = A(x) B(t)$$

$$T(0, t) = A(0) B(t)$$

$$T(L, t) = A(L) B(t)$$

} Boundary conditions $A(0) = A(L) = 0$

trivial solution, $B(t) = 0$ for all $T \Rightarrow T(x, t) = 0$

$\Rightarrow A(0) = 0$ and $A(L) = 0$ must hold

→ found IVP for space ODE

- Solve space ODE

$$\frac{d^2 A}{dx^2} + m A = 0$$

$$A(0) = 0$$

$$A(L) = 0$$

Problem: we do not know $m \Rightarrow$ two cases

$$m = 0$$

$$\frac{d^2 A}{dx^2} = 0 \Rightarrow$$

$$A(x) = c_1 + c_2 x$$

$$A(0) = 0 \Rightarrow c_1 = 0$$

$$A(L) = 0 \Rightarrow c_2 L = 0 \Rightarrow c_2 = 0 \left. \vphantom{\begin{matrix} A(0) = 0 \\ A(L) = 0 \end{matrix}} \right\} \text{trivial solution}$$

$$m \neq 0 \quad \frac{d^2 A}{dx^2} + m A = 0$$

characteristic equation

$$\lambda^2 + m = 0 \quad \Rightarrow \quad \lambda_{1/2} = \pm \sqrt{-m}$$

Solution: $A(x) = c_1 \exp(\lambda_1 x) + c_2 \exp(\lambda_2 x)$

$$m < 0: \quad A(x) = c_1 + c_2 = 0 \quad \Rightarrow \quad c_1 = -c_2$$

$$A(L) = c_1 \exp(\lambda_1 L) + c_2 \exp(\lambda_2 L) = 0$$

$$\Rightarrow \quad \frac{c_1}{c_2} \exp(\lambda_1 L) = -\exp(\lambda_2 L)$$

$$\Rightarrow \quad -\exp(\lambda_1 L) = -\exp(\lambda_2 L)$$

$$\Rightarrow \quad \lambda_1 = \lambda_2 \quad \Rightarrow \quad \text{only possible for } m = 0$$

$$m > 0$$

$$\Rightarrow z_{1,2} = \pm \sqrt{m} i = 0 \pm \sqrt{m} i$$

$$a \neq b i$$

$$A(x) = \exp(ax) (c_1 \cos(bx) + c_2 \sin(bx))$$

$$A(x) = c_1 \cos(\sqrt{m} x) + c_2 \sin(\sqrt{m} x)$$

$$A(0) = 0 \Rightarrow c_1 = 0$$

$$A(L) = 0 \Rightarrow c_2 \sin(\sqrt{m} L) = 0$$

$$\text{if } c_2 \neq 0 \Rightarrow \sin(\sqrt{m} L) = 0$$

infinite solutions: $\sin(x) = 0$ for $0, \pi, 2\pi, \dots, n\pi$

$$\sqrt{m} L = n\pi \Rightarrow m_n = \left(\frac{n\pi}{L}\right)^2 \quad n = 1, 2, \dots$$

$$A_n(x) = c_2 \sin\left(\frac{n\pi}{L} x\right)$$

↑
still unknown

- Solve time equation

$$\frac{dB}{dt} = -\alpha m_n B$$

$$m_n > 0 = \left(\frac{n\pi}{L}\right)^2$$

$$B(x,t) = C_1 \exp(-\alpha \left(\frac{n\pi}{L}\right)^2 t)$$

- PDE solution

$$T(x,t) = A(x)B(t)$$

$$T(x,t) = C \sin\left(\frac{n\pi}{L}x\right) \exp\left(-\alpha \left(\frac{n\pi}{L}\right)^2 t\right) \quad \omega = \frac{n\pi}{L}$$

$$T(x,t) = C \sin(\omega x) \exp(-\alpha \omega^2 t)$$

$$\text{IVP} \quad T(x,0) = f(x) = C \sin(\omega x)$$

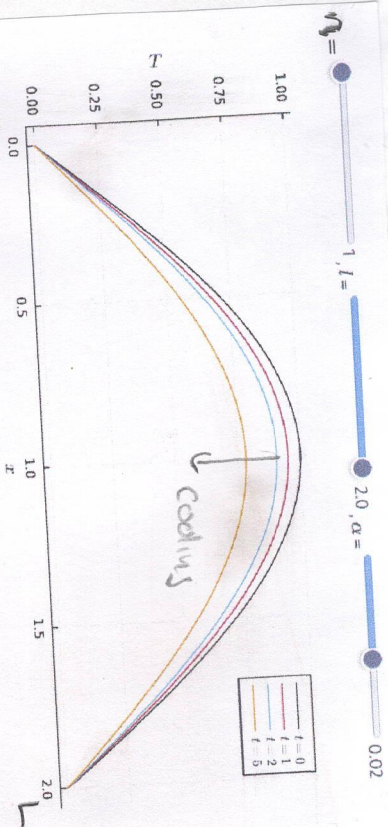
$$= C \sin\left(\frac{n\pi}{L}x\right)$$

$$\text{original } f(x) = \sin\left(\frac{\pi}{L}x\right)$$

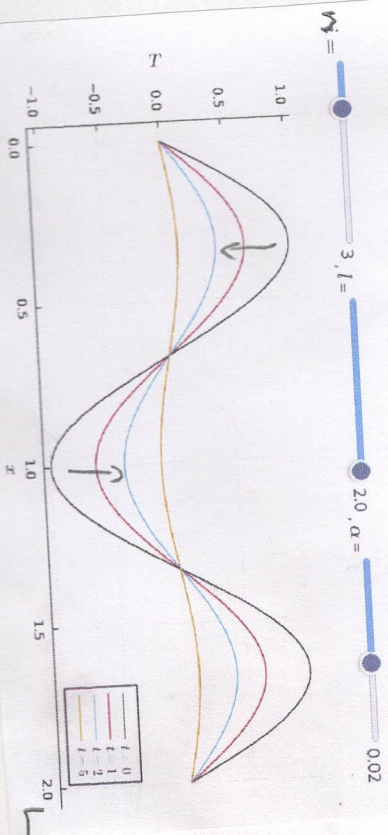
$$\Rightarrow C = 1$$

$$n=1$$

$$T(x,t) = \sin\left(\frac{\pi}{L}x\right) \exp\left(-\alpha \left(\frac{\pi}{L}\right)^2 t\right)$$



$$\alpha \uparrow \Rightarrow \frac{\partial T}{\partial t} \uparrow$$



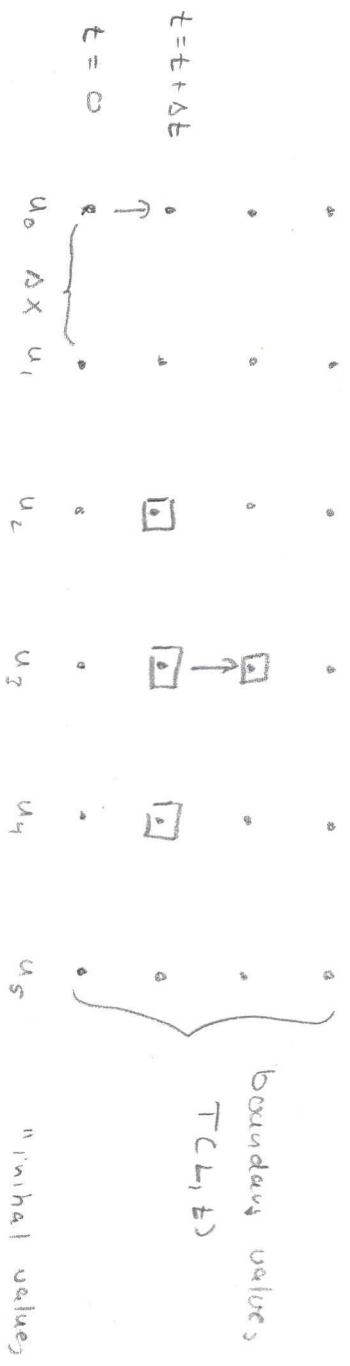
$$T(x,t) = \sin\left(\frac{3\pi}{L}x\right) \exp\left(-\alpha\left(\frac{3\pi}{L}\right)^2 t\right)$$

Numerical Solution to PDEs

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

"Method of Lines": RHS as finite difference approximation

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$



$$du_0 = 0 = u_0(t+\Delta t)$$

$$du_1 = u_1(t+\Delta t) = \alpha \frac{u_2 - 2u_1 + u_0}{\Delta x^2} = \frac{\alpha}{\Delta x^2} (u_2 - 2u_1 + u_0)$$

$$\vec{du} = \frac{\partial}{\Delta x^2} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ du_0 & -2 & 1 & 0 & 0 & 0 \\ du_1 & 1 & -2 & 1 & 0 & 0 \\ du_2 & 0 & 1 & -2 & 1 & 0 \\ du_3 & 0 & 0 & 1 & -2 & 1 \\ du_4 & 0 & 0 & 0 & 1 & -2 \\ du_5 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix}$$

$$\vec{du} = \frac{\partial}{\Delta x^2} A \vec{u} \quad \Delta x^2 = \frac{L}{n-1}$$

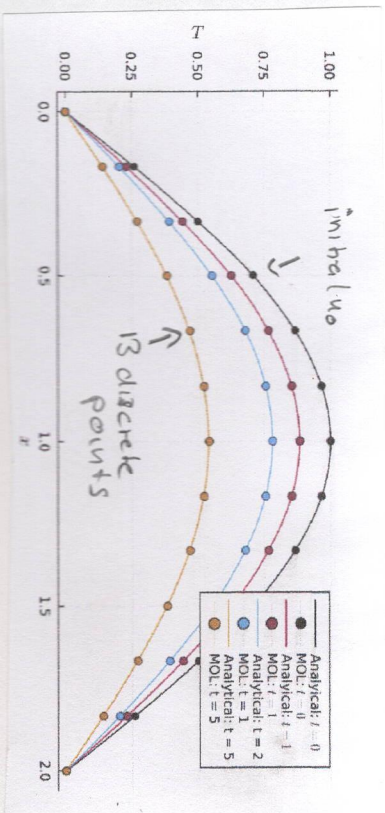
$$A \in \mathbb{R}^{n \times n}$$

System of first order ODEs $\Rightarrow$ use ODE solver

$$\left. \begin{array}{l} du_0 = 0 \\ du_{n-1} = 0 \end{array} \right\} \text{Dirichlet Boundary Condition}$$

- can take on other values than zero
- picked zero to compare to analytical solution

Num en cel Solution



function $f(u, p, t)$

$$\alpha = p[1]$$

$$\Delta x = p[2]$$

$$A = p[3]$$

$$du = \frac{\alpha}{\Delta x^2} A u$$

return du

end

n is length of u

and A is size (n, n)

Other Boundary Conditions



Dirichlet: fixed value

$$\Rightarrow \frac{du_0}{dt} = \frac{du_L}{dt} = 0$$

Neumann: fixed gradient / flux

$$\frac{du_0}{dx} = \frac{du_L}{dx} = C$$

constant

often $C=0$: Zero flux $\Rightarrow$ "adiabatic"



$$\frac{u_1 - u_{-1}}{2\Delta x} = C \Rightarrow u_{-1} = u_1 - 2\Delta x \cdot C$$

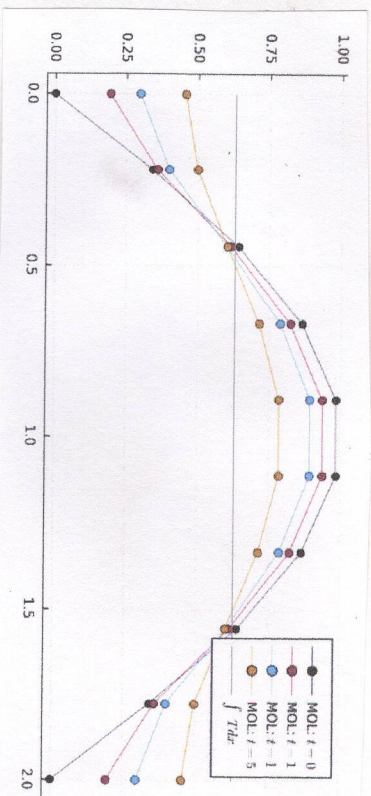
$$\partial u_0 = u(t + \Delta t) = \alpha \frac{u_1 - 2u_0 + u_{-1}}{\Delta x^2} = \frac{\alpha}{\Delta x^2} (u_1 - 2u_0 + u_{-1} - 2\Delta x \cdot C)$$

for $c=0$

$$du_0 = \frac{\partial}{\partial x^2} (2u_1 - 2u_0)$$

$$A = \begin{bmatrix} -2 & 2 & 0 & 0 & \dots & \dots & 0 \\ 0 & -2 & 1 & 0 & \dots & \dots & 0 \\ 0 & 0 & -2 & 1 & \dots & \dots & 0 \\ & & & & \ddots & & \\ & & & & & -2 & 2 \end{bmatrix}$$

$$d\vec{u} = \frac{\partial}{\partial x^2} A \vec{u}$$



Numerical Solution Stability

Integration through Euler's Method

$u_i^j \leftarrow \begin{matrix} \text{time} \\ \text{space} \end{matrix}$

$$du = \frac{u_i^{j+1} - u_i^j}{\Delta t} = \alpha \frac{u_{i+1}^j - 2u_i^j + u_{i-1}^j}{\Delta x^2}$$

- could use this to get u_i^{j+1}

$$\text{system is stable if } \frac{2\alpha\Delta t}{\Delta x^2} \leq 1$$

"CFL condition"
(Courant-Friedrichs-Lewy)

$\Rightarrow \Delta t$ must be sufficiently small

$$\text{note } \tau_{\text{cond}} = \frac{\Delta x^2}{\alpha} \quad \text{time scale of conduction}$$

$$\Rightarrow \Delta t < \frac{1}{2} \tau_{\text{cond}}$$

$\Rightarrow$ may need small Δt and solution is slow

2D case

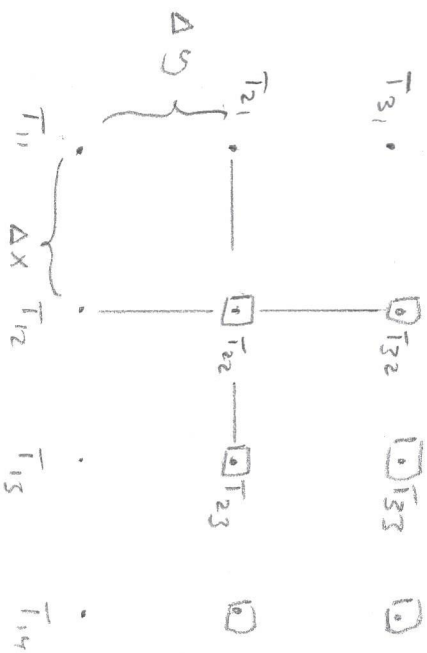
$$= \delta \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$



$$\frac{70}{2110}$$

$$\frac{\overline{T}_{i+1, i} - 2\overline{T}_{i, i} + \overline{T}_{i-1, i}}{\Delta x^2} + \frac{\overline{T}_{i+1, i} - 2\overline{T}_{i, i} + \overline{T}_{i-1, i}}{\Delta y^2}$$

boundary
nodes



$\downarrow$

Raft Dispersion

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}$$

$$C(x,0) = H\delta(x)$$

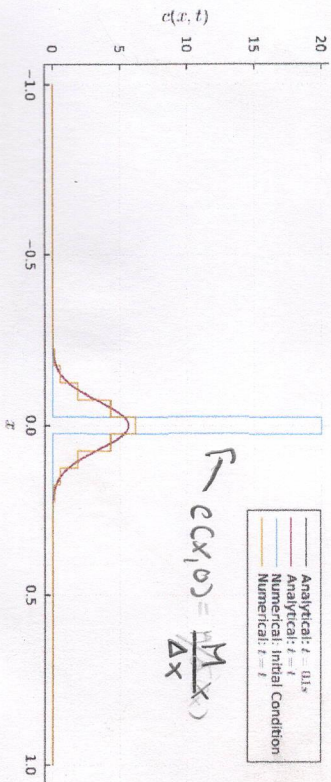
$$C(x,t) \rightarrow 0 \quad x \rightarrow \infty$$

solution identical to heat problem

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

$$C(x,t) = \frac{H}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

$$C(x,0) = \frac{H}{\Delta x} \delta(x)$$

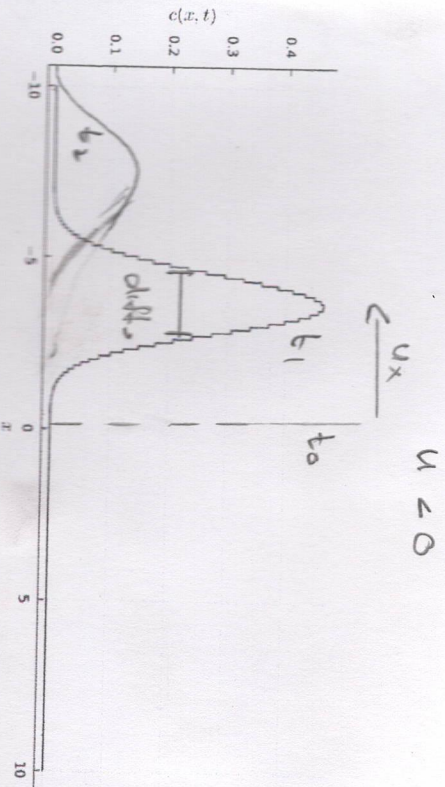


Puff Dispersion + Advection

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} - u_x \frac{\partial c}{\partial x}$$

$$\text{MAC: } \frac{\partial c_i}{\partial t} = D \left(\frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta x^2} \right) - u_x \left(\frac{C_{i+1} - C_{i-1}}{2 \Delta x} \right)$$

Forward time
centered difference
(FTSC)



- even for good CFL condition
Solution will develop oscillations
 - need more sophisticated
scheme for advection problem
 - > upwind scheme / Trueman's
method
 - > better to solve C_i through
loop instead of ODE
- Solver